

SEFM'26 Artefact Evaluation Guidelines

The artefact evaluation (AE) differs according to the type of paper submitted:

- Accepted regular papers *may* be invited to submit an accompanying artefact for evaluation by the AE committee.
- Tool or artefact papers *must* submit an artefact that *at least* satisfies the requirements for the functional badge (see 1 in tbl. 1). Papers not meeting this clause are rejected.

1 Badge claims

Authors should indicate the [EAPLS](#) badges they intend to claim with justification why their artefact qualifies for the claimed badges. Tbl. 1 details the badges available. Papers may claim *up to* two badges.

Example 1 (Claiming badges). We claim two badges (i) **Artefacts Available** and (ii) **Artefacts (Functional and) Reusable**. The reasons why our tool fulfils the requirements set out by the EAPLS scheme for are outlined below. ■

2 Submission format

When packaging their artefact for submission, authors should provide specific scenarios describing how it can be concretely used. Authors can also consider variations on those scenarios to showcase the robustness or reusability of their artefact. Sec. 3 describes the template authors should follow when structuring their AE submissions.

Every submitted artefact must include the following:

- The URL from where the artefact can be downloaded.
- A quick start guide explaining how the artefact environment is set up.

The AE committee expects artefacts to be executable on major operating systems (macOS, Linux, Windows). Authors should include both

- A Docker image or OVA/OVF virtual machine (VM) containing the artefact;
and
- A separate archive containing the artefact source files.

The Docker image or VM enables reviewers to execute and evaluate the artefact without installing additional software on their machines. The source archive allows reviewers who are familiar with the underlying language or platform to run the artefact directly from source.

1 Available



To claim the *available* badge, authors must provide a link to a publicly accessible repository that permanently hosts artefacts. If this is not possible at the time of submission, authors should indicate the repository on which they intend to host the artefact (e.g. [Zenodo](#), [Software Heritage](#)). The hosting platform should not track the identities of downloaders: using non-compliant hosting platforms is valid grounds for outright artefact rejection.

2.1 Functional



To claim the *functional* badge, the submitted artefact should be reasonably documented and include evidence of proper validation. Authors should identify all the individual outcomes in the paper, which must be reproducible through the artefact. Each functional outcome should be numbered, $F_1, \dots, F_n$, and accompanied by a short explanation that links that outcome obtained via the artefact to the one in the paper.

Example 2 (Functional outcomes).

F_1 Experiment 4 shows that our monitoring approach is efficient in terms of the memory consumed.^(Outcome) The memory consumption printed on the terminal by this experiment corresponds to rows 1-4 listed in Tbl. 3.^(Link with paper) ■

2.2 (Functional and) Reusable



To claim the *reusable* badge, the submitted artefact should be of high quality that exceeds the requirements for the functional badge. Authors should outline individual reusability claims, $R_1, \dots, R_n$, that demonstrate how the artefact may be used in scenarios beyond those covered in the paper. Each scenario should be presented as an example that AE reviewers can replicate.

Example 3 (Reusability claims).

R_1 Our first example shows how the monitoring software detects invalid handshake protocols besides the ones studied in Sec. 4 of the paper.^(Claim) These instructions set-up the system and launch the monitors, which flag a protocol violation in step 4, after the first event.^(Replicable scenario) ■

OR

Tbl. 1: EAPLS badges that can be claimed by artefact authors

3 Evaluation instructions

The AE instructions should be attached as a post-bibliography appendix in the original paper. It must detail the steps reviewers should follow to confirm the functionality or reusability claims of tbl. 1. This appendix is reserved for the AE procedure and will *not* be published with the camera-ready version of the paper. We recommend the appendix format outlined in secs. A.1 to A.4.

A.1 Badge claims

Authors should state the EAPLS badges they intend to claim, motivating why the submitted artefact qualifies for each badge. Tool papers *must* claim the functional badge and meet its requirements. Refer to tbl. 1 for examples on how to specify the functional outcomes and reusability claims. Functional outcomes and reusability claims can be organised under these sub-headings.

Functional outcomes The following functional outcomes are expected.

F₁ This outcome...

Reusability claims The following reusability claims are covered.

R₁ This claim...

A.2 Quick start

This phase confirms whether the artefact and its environment are fully operational prior to the main evaluation. The quick start guide should consist of a *short* sequence of sanity checks that reviewers execute. Ideally, authors provide a script that automates these checks. Authors are advised to give a high-level directory structure of their submitted artefact since this helps AE reviewers familiarise themselves with the artefact.

Example 4 (Directory structure). Fig. 1 describes the directory layout of our submitted artefact. The files used in secs. A.3 and A.4 are underlined. AE reviewers may safely ignore the other directories. ■

The quick start guide aims to facilitate the *kick-the-tyres* phase of the AE. At this stage, AE reviewers may communicate with artefact authors to give early feedback or ask for further instructions in case of difficulties.

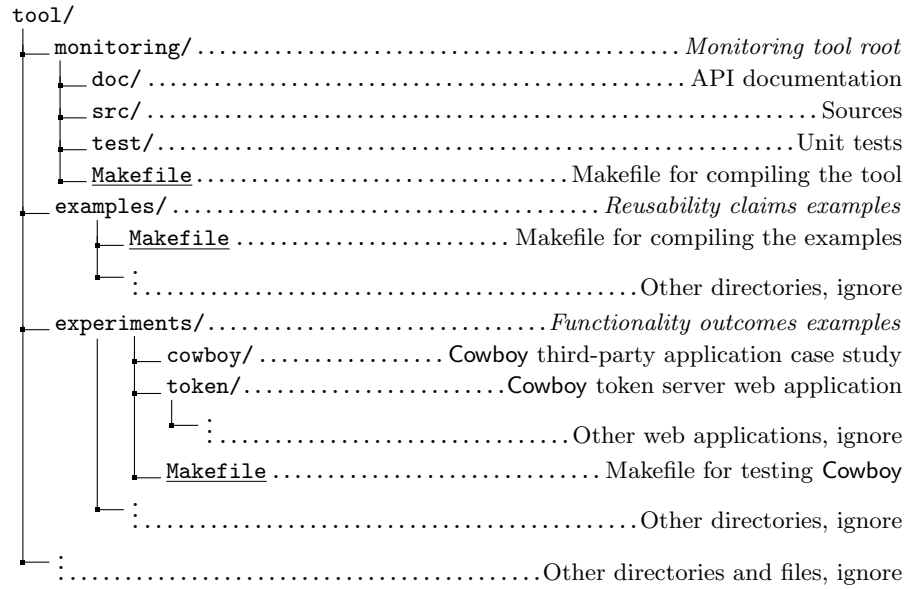


Fig. 1: Tool directory structure example

A.3 Functional evaluation

This evaluation replicates the functional outcomes $F_1, \dots, F_n$ listed in sec. A.1. Authors should structure this appendix section as a sequence of steps AE reviewers follow to verify all the functional outcomes. Reference to individual functional outcomes should be made by authors when arguing why or how each outcome is met. Justification should be given whenever the results obtained through the artefact differ from those in the paper.

Example 5 (Justifying difference in results). The rendered plot produced by experiment 5 (functional outcome F_5) reflects the same trends of Fig. 7 in the paper, albeit on a smaller scale. This difference between the artefact plot and Fig. 7 results from the *lower* load we use for our artefact set-up to keep the execution time reasonable, per the AE guidelines. ■

A.4 Reusability evaluation

This evaluation validates the reusability claims $R_1, \dots, R_n$ identified in sec. A.1. Authors should structure this section of the appendix as a sequence of steps AE reviewers follow to explore the reusability of the submitted artefact. These steps can (i) direct AE reviewers to inspect content packaged with the artefact (*e.g.* documentation), (ii) give scenarios where the artefact may be reused (*e.g.* imported as a library into other software), or (iii) follow instructions which showcase how the artefact can be used out-of-the-box to tackle use cases besides those given in the paper (*e.g.* a short tutorial with examples).

Example 6 (Reusing the artefact on other examples). Our main monitoring module accepts command line switches that enable the software to be used as a stand-alone tool (reusability claim R_2). This tutorial shows how an arbitrary client-server system written in Erlang can be monitored by launching the monitoring module using the `-standalone` and `-mfa` switches as follows... ■

AE reviewers should be able to complete the evaluation in ≈ 4 hours. If the evaluation relies on long-running experiments (*e.g.* to gather empirical data), the evaluation harness should provide a push-button method for executing experiments with *minimal* intervention from AE reviewers (*e.g.* using a script). Artefacts packaged as Docker images or OVA/OVF virtual machines reduce the AE set-up overhead.